SeaLLMs 3: Open Foundation and Chat Multilingual Large Language Models for Southeast Asian Languages

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Project page: https://seallms.github.io/

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Abstract

Large Language Models (LLMs) have shown remarkable abilities across various tasks, yet their development has predominantly centered on high-resource languages like English and Chinese, leaving low-resource languages underserved. To address this disparity, we present SeaLLMs 3, the latest iteration of the SeaLLMs model family, tailored for Southeast Asian languages. This region, characterized by its rich linguistic diversity, has lacked adequate language technology support. SeaLLMs 3 aims to bridge this gap by covering a comprehensive range of languages spoken in this region, including English, Chinese, Indonesian, Vietnamese, Thai, Tagalog, Malay, Burmese, Khmer, Lao, Tamil, and Javanese. Leveraging efficient language enhancement techniques and a specially constructed instruction tuning dataset, SeaLLMs 3 significantly reduces training costs while maintaining high performance and versatility. Our model excels in tasks such as world knowledge, mathematical reasoning, translation, and instruction following, achieving state-of-the-art performance among similarly sized models. Additionally, we prioritized safety and reliability by addressing both general and culture-specific considerations and incorporated mechanisms to reduce hallucinations. This work underscores the importance of inclusive AI, showing that advanced LLM capabilities can benefit underserved linguistic and cultural communities.

1 Introduction

Large Language Models (LLMs) such as GPT-4 (OpenAI, 2023) and Gemini (Anil et al., 2023) have demonstrated remarkable capabilities across a wide array of tasks, ranging from natural language understanding and generation to more specialized domain applications (Zhao et al., 2023). These models have proven valuable, offering substantial benefits to the global community, especially through the proliferation of open-source LLMs such as Llama (Touvron et al., 2023a,b), Mistral (Jiang et al., 2023), Qwen (Bai et al., 2023; Yang et al., 2024), and Gemma (Mesnard et al., 2024). However, the majority of these efforts have been concentrated on high-resource languages such as English and Chinese, or well-developed regions like Europe (Zhang et al., 2023; Ahuja et al., 2023). Consequently, the development of LLMs tailored for low-resource languages or underdeveloped regions has been significantly overlooked, resulting in a lack of inclusivity and equitable distribution of AI advancements across diverse linguistic and cultural communities (Qin et al., 2024; Huang et al., 2024; Liu et al., 2024).

To bridge this gap, we introduced the SeaLLMs model (Nguyen et al., 2023c), specifically designed LLMs for Southeast Asian languages. Southeast Asia (SEA) is a region with a rich diversity of languages spoken by millions of people, yet it suffers from a significant lack of language technology support (Aji et al., 2022). The initiative of SeaLLMs thus aims to make the benefits of LLMs accessible to speakers of these languages, addressing their unique linguistic and cultural nuances. Following this endeavor, several other models have been dedicated to this region as well, such as SEA-LION (AI Singapore, 2023) and Sailor (Dou et al., 2024). However, these models often face significant limitations: they are typically released only as foundational or chat models, offer limited options in terms of model size, and cover a limited number of SEA languages. Moreover, the relatively scarce availability of language corpora further constrains the amount of training data available, hindering the development and performance of these models.

In this work, we introduce **SeaLLMs 3**, the latest iteration of the SeaLLMs model family. This version is designed to cover a more diverse ar-

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ray of Southeast Asian languages, including English, Chinese, Indonesian, Vietnamese, Thai, Tagalog, Malay, Burmese, Khmer, Lao, Tamil, and Javanese. Different from the conventional continue-pretraining paradigm (Zhao et al., 2024a; Nguyen et al., 2023c; Dou et al., 2024), we conduct efficient language enhancement by training language-specific neurons only based on a foundation model (Zhao et al., 2024b), significantly reducing the overall training cost. Moreover, such targeted training also ensures that the performance of high-resource languages can remain unaffected during the enhancement. Furthermore, SeaLLMs 3 is trained using a specially constructed instruction tuning dataset that encompasses a wide variety of task types and carefully balanced language distributions. This approach ensures that the model can handle the linguistic diversity of the Southeast Asian region while maintaining high performance and versatility across different applications. As a result, it achieves state-of-the-art performance among models with similar sizes, excelling across a diverse array of tasks such as world knowledge (Zhang et al., 2023), mathematical reasoning (Shi et al., 2023), translation (Costa-jussà et al., 2022), and instruction following.

In the meantime, we pay special attention to the model's reliability and trustworthiness during its development, which are often under-considered in multilingual settings (Deng et al., 2023). In particular, we address both general and culture-specific safety considerations to ensure the models provide contextually appropriate responses. The model is also specifically trained to be aware of its knowledge boundary and refuse what it does not know. To evaluate this capability, we introduce a novel benchmark, SeaRefuse, which assesses the ability of LLMs to refuse questions beyond their knowledge boundaries. This focus on safety and reliability has resulted in SeaLLMs 3 exhibiting reduced hallucination and delivering safe, coherent responses, especially in queries closely related to Southeast Asian culture.

We open-source both the foundational and chat models of SeaLLMs 3[1]. The foundational model could serve as a base for conducting instruction tuning tailored to specific application requirements. Meanwhile, the chat model has already undergone instruction tuning and is ready for direct use of handling a wide range of tasks.

[1] https://huggingface.co/collections/SeaLLMs/seallms-v3-668f3a52e1e6fbaad5752cdb

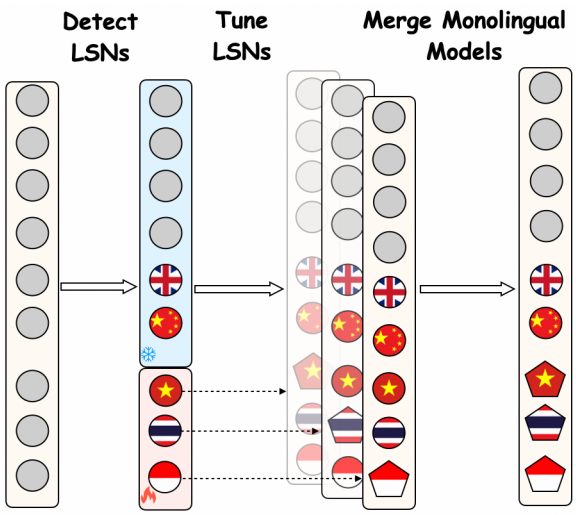

Figure 1: Language-Specific Neuron Training.

2 Pre-training

2.1 Pre-training Data

Building on the efforts from previous versions of SeaLLMs (Nguyen et al., 2023c), we have incorporated corpora from a wider range of data sources to enhance diversity. Specifically, we have integrated fundamental knowledge from Wikipedia (Foundation) and textbooks (Ben Allal et al., 2024), journalistic materials such as CC-News (Crawl), web-based corpora from CulturaX (Nguyen et al., 2023a), and MADLAD-400 (Kudugunta et al., 2023). We have also improved the data processing pipeline including the language model filtering and duplicate removal to improve the data quality.

Additionally, we explore the utilization of model-synthetic data for training, which received much attention recently (Ben Allal et al., 2024). Starting with manual annotation of domain-specific knowledge points in SEA languages, we then employed stronger models to generate targeted tutorial-style content, thereby enhancing SeaLLMs 3 with enriched regional knowledge in a more explicit form.

2.2 Language-Specific Neuron Training

We built our model based on the Qwen2 model family (Yang et al., 2024) and further conducted language enhancement to augment its capability in SEA languages. This approach allows the model to quickly inherit foundational knowledge from Qwen, rather than learning it from scratch.

The most straightforward method for language enhancement is typically through continued pertain-

ing (Zhao et al., 2024a), which we also used for previous versions of SeaLLMs. However, as discussed, the relatively scarce availability of language corpora limits the amount of training data, hindering the development and performance of these models. Furthermore, it is often observed that such direct continued pretraining can compromise the model's original capacity in high-resource languages like English and Chinese (Dou et al., 2024).

In this iteration, we adopt Language-Specific Neuron (LSN) training for efficient language enhancement, as shown in Figure 1. Recent studies have found that certain language-specific neurons in language models are responsible for processing specific languages. For instance, Zhao et al. (2024b) discovered that language-specific neurons comprise only about 0.1% of all parameters. Thus, the capabilities of a language can be enhanced by training its corresponding LSNs while preserving the multilingual abilities of other languages. To efficiently train SeaLLMs 3, we employ the parallel language-specific neuron detection method proposed by Zhao et al. (2024b). As shown in the left most part of Figure 1, this method allows us to identify the LSNs of SEA languages using language-specific training data, selected as a downsampled subset of the corpora from the training data. We then specifically train these detected LSNs to develop multiple monolingual LLMs in SEA languages, which are subsequently merged to create a unified multilingual LLM for SEA languages. Additionally, to maintain proficiency in English and Chinese from the original foundation model, we detect their respective LSNs and exclude them from the entire pre-training process.

This method offers several advantages. First, it requires relatively less training data since the training is more targeted, which significantly reduces training costs. Second, because the training is targeted, we can ensure that the performance of high-resource languages from the original foundation model remains unaffected. LSNs operate independently and do not influence one another, avoiding the sacrifices seen with previous methods.

3 Supervised Fine-Tuning (SFT)

3.1 Supervised Fine-tuning Data

Most existing open-source supervised fine-tuning (SFT) datasets are predominantly in English (Wei et al., 2022; Taori et al., 2023), which presents a challenge for developing effective models for

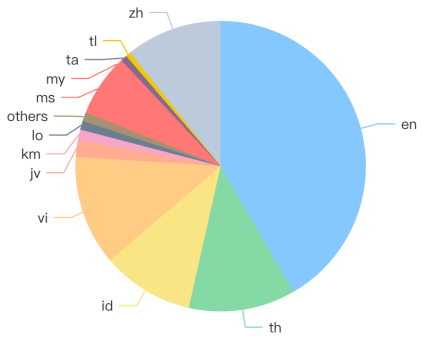

Figure 2: Language distribution of the SFT data

Southeast Asian languages. To address this, we employed various techniques to construct our SFT pool. For example, we selectively translate some high-quality English data to SEA languages with quality filtering, conduct self-instruction to automatically generate SFT data of certain types, and use various prompting strategies (Madaan et al., 2023; Nguyen et al., 2023b). Following our previous practice, native speakers have been actively engaged throughout the entire SFT data construction process. They manually collect and write seed questions and topic lists, ensuring linguistic and cultural accuracy from the outset. Additionally, native speakers verify, filter, and edit the synthetic SFT data to maintain high quality.

Our preliminary experiments indicated that relying heavily on dominant English data adversely affects performance. To mitigate this, we strive to maintain a relatively good balance of language representation in our training data this time. Figure 2 shows the language distribution of our SFT data. While English remains a significant portion of the dataset, substantial representation is given to other Southeast Asian languages such as Indonesian, Vietnamese, Thai, and others, ensuring a comprehensive and diverse linguistic foundation for the model's training.

Since the first release of SeaLLMs (Nguyen et al., 2023c), the task types of SFT data have been significantly expanded. The dataset now includes a diverse range of task types such as coding, math, education-related content, reasoning, general dialogue, table-related tasks, open-domain QA, and many more. This expansion ensures that the model is well-rounded and capable of handling a variety of queries and tasks. Additionally, SFT with multiple turns has been significantly increased to enhance the model's ability to engage in natural,

multi-turn dialogues, improving its conversational fluency and coherence.

Model safety, trustworthiness, and reliability are also important factors for constructing the SFT pool. To address this, we specifically constructed refusal-type data, enabling the model to decline questions beyond its knowledge boundaries, such as those involving non-existing entities. Furthermore, we carefully curated safety-related data, including both general safety data (which are culturally independent, such as general moral principles) and country-specific safety data (which are culturally sensitive). This approach ensures that the model can be safely deployed with cultural considerations in mind, providing accurate and appropriate responses across different cultural contexts.

3.2 Training Details

Two stages of training are employed to optimize the model's performance. In the first stage, a large volume of SFT data is used to equip the model with instruction-following capabilities and to familiarize it with different task types. In the second stage, a smaller but high-quality SFT dataset is utilized to fine-tune the model, ensuring it performs exceptionally well on important tasks.

During the training process, different samples are packed together for efficiency, with a maximum length set at 8,192 tokens. The learning rate is set at 1.0e-5, with a warmup ratio of 0.1. Additionally, gradients are clipped at a maximum of 1.0 to prevent exploding gradients.

4 Evaluations

We conduct extensive evaluations against models with similar sizes, including Sailor-7B / Sailor-7B-Chat (Dou et al., 2024), Gemma-7b / Gemma-7b-it (Mesnard et al., 2024), Qwen2-7B / Qwen2-7B-Chat (Yang et al., 2024), Meta-Llama-3-8B / Meta-Llama-3-8B-Instruct (Touvron et al., 2023b), Aya-23-8B (Aryabumi et al., 2024), and the previous versions (mainly v2.5) of the SeaLLMs (Nguyen et al., 2023c). The models are listed by their release date in the following result tables.

The evaluations can be generally categorized into the following two dimensions:

- **Model Capability**: We assess the model's performance on human exam questions, its proficiency in mathematics, its ability to follow multi-turn instructions, and its translation accuracy of different language pairs.
- **Model Trustworthiness**: We evaluate the model's safety and tendency to hallucinate, particularly in the context of Southeast Asia.

4.1 Model Capability

4.1.1 Multilingual World Knowledge

Dataset We utilized the M3Exam dataset (Zhang et al., 2023), comprising real human exam questions collected from different countries and spanning different subjects and educational stages. This dataset effectively tests the model's multilingual world knowledge in a manner more akin to real-world settings. We take the questions in English (en), Chinese (zh), Indonesian (id), Vietnamese (vi), and Thai (th). We also employ the translated MMLU (Hendrycks et al., 2021) questions for evaluation, which primarily tests the cross-lingual alignment of the model as the required knowledge is still mainly Western-focused. For evaluation, we employ the SeaExam toolkit[2] and measured performance using accuracy as the metric.

Results As shown in Table 1 for results on M3Exam dataset, our models, SeaLLMs-v3-7B and SeaLLMs-v3-7B-Chat, demonstrate competitive performance, with SeaLLMs-v3-7B-Chat achieving the highest average score (0.692) and the highest average score for SEA languages (0.592). Compared to the previous version, SeaLLMs-7B-v2.5, our latest models show significant improvement in overall performance specifically in handling Southeast Asian languages. Furthermore, while the Qwen2-7B-Instruct model performs exceptionally well in English and Chinese, our models exhibit superior performance across a broader range of Southeast Asian languages, highlighting their enhanced multilingual capabilities.

Table 2 shows the results of different models on the translated MMLU dataset, we can see that our SeaLLMs-v3-7B and SeaLLMs-v3-7B-Chat models also outperform other models, particularly in Southeast Asian languages. Compared to the previous SeaLLMs-7B-v2.5 version, our latest models show substantial improvements, particularly in handling Southeast Asian languages (with avg_sea from 52.4 to 58.2).

4.1.2 Multilingual Math

Dataset We assess the multilingual mathematics capabilities using the MGSM dataset (Shi et al.,

[2] https://github.com/DAMO-NLP-SG/SeaExam

Model	en	zh	id	th	vi	avg	avg_sea
Gemma-7B	73.2	51.9	47.5	46.0	59.4	55.6	51.0
Sailor-7B-Chat	66.0	65.2	47.5	46.2	51.3	55.2	48.3
SeaLLM-7B-v2.5	75.8	58.1	49.9	50.2	62.2	59.2	54.1
Sailor-14B	74.8	84.0	53.6	52.8	62.1	65.5	56.2
Sailor-14B-Chat	74.9	84.3	55.3	56.6	63.7	67.0	58.5
Qwen2-7B	**81.5**	87.4	53.0	47.9	62.8	66.5	54.6
Qwen2-7B-Instruct	80.9	**88.0**	55.8	55.5	62.4	68.5	57.9
SeaLLMs-v3-7B	80.9	86.3	54.5	53.0	62.8	67.5	56.8
SeaLLMs-v3-7B-Chat	80.9	87.4	**55.8**	**56.9**	**64.9**	**69.2**	**59.2**

Table 1: Results of multilingual world knowledge with the M3Exam benchmark

Model	en	zh	id	th	vi	avg	avg_sea
Gemma-7B	63.4	50.9	54.5	49.0	49.4	53.5	51.0
Sailor-7B-Chat	55.8	47.2	48.4	41.4	46.2	47.8	45.4
SeaLLM-7B-v2.5	65.2	54.4	56.5	47.9	52.8	55.3	52.4
Sailor-14B	61.8	56.4	57.0	48.2	53.5	55.4	52.9
Sailor-14B-Chat	62.7	56.1	56.7	49.6	54.1	55.8	53.5
Qwen2-7B	71.0	64.2	60.2	52.0	56.6	60.8	56.3
Qwen2-7B-Instruct	70.8	63.5	59.9	52.4	56.8	60.7	56.4
SeaLLMs-v3-7B	70.6	**65.4**	61.7	53.6	**58.7**	62.0	58.0
SeaLLMs-v3-7B-Chat	**71.3**	64.7	**62.5**	**54.4**	57.8	**62.2**	**58.2**

Table 2: Results of multilingual world knowledge with the translated MMLU benchmark

2023). Originally, MGSM comprises testing samples solely in English, Chinese and Thai. To extend this dataset to other SEA languages, specifically Indonesian, Malay, and Vietnamese, we utilize Google Translate to translate the original English questions to those questions. It is important to note that in our translations, we adhere to the numerical notation conventions of each respective country. For instance, in both Indonesian and Vietnamese, dots are used as thousands separators, and commas as decimal separators, which is the reverse of the English numeral system. We follow the same convention when evaluating the model's generations.

Results Table 3 presents the evaluation results on the MGSM benchmark, under both the few-shot setting (for testing base versions) and the zero-shot setting (for testing chat versions). In the few-shot setting, SeaLLMs-v3-7B demonstrates the highest average score (63.1), outperforming other models such as Qwen2-7B (62.3) and GLM-4-9b (60.2), particularly excelling in Indonesian and Thai. In the zero-shot setting, the chat version of SeaLLMs-v3-7B-Chat achieves the highest average score (73.1), showing strong performance across all languages. This highlights SeaLLMs-v3-7B-Chat's superior adaptability and robustness in multilingual math tasks compared to its counterparts like Qwen2-7B-instruct (68.4).

4.1.3 Multilingual Instruction-following

Dataset As there is no publicly available dataset for testing the model's multi-turn instruction-following capability in SEA languages, we construct our own benchmark, namely SeaBench[3], for such evaluation. SeaBench consists of multi-turn human instructions spanning various task types for Indonesian, Vietnamese, and Thai. Following MT-Bench (Zheng et al., 2023), we consider 8 task types including writing, roleplay, extraction, reasoning, math, coding, knowledge I (STEM), and knowledge II (humanities/social science). Additionally, considering the characteristics of the multilingual setting, we include two more task types: safety and life. The safety task tests whether the model will respond to unsafe queries in a local context, while the life task includes questions likely

[3]It will be publicly available at https://huggingface.co/datasets/SeaLLMs/SeaBench

MGSM	en	id	ms	th	vi	zh	avg
Few-shot setting							
Gemma-7B	64.8	41.2	43.2	38.0	34.0	39.6	43.5
Sailor-7B	34.4	25.2	22.8	24.8	22.4	26.4	26.0
Meta-Llama-3-8B	56.8	36.0	33.6	34.8	33.6	43.6	39.7
GLM-4-9B	78.0	53.6	**57.2**	46.0	**56.8**	69.6	60.2
Qwen2-7B	**79.6**	58.8	56.8	54.8	54.8	69.2	62.3
SeaLLMs-v3-7B	78.8	**59.2**	56.8	**56.8**	54.8	**72.0**	**63.1**
Zero-shot setting							
Gemma-1.1-7B-it	58.8	32.4	34.8	31.2	39.6	35.2	38.7
Sailor-7B-Chat	33.6	22.4	22.4	21.6	25.2	29.2	25.7
SeaLLM-7B-v2.5	79.6	69.2	**70.8**	61.2	66.8	62.4	68.3
Meta-Llama-3-8B-Instruct	77.6	48.0	57.6	56.0	46.8	58.8	57.5
GLM-4-9B-Chat	72.8	53.6	53.6	34.8	52.4	70.8	56.3
Qwen2-7B-Instruct	**82.0**	66.4	62.4	58.4	64.4	76.8	68.4
SeaLLMs-v3-7B-Chat	74.8	**71.2**	**70.8**	**71.2**	**71.2**	**79.6**	**73.1**

Table 3: Results of multilingual math with the MGSM benchmark.

Model	id			th			vi			avg
	turn1	turn2	avg	turn1	turn2	avg	turn1	turn2	avg	
Sailor-7B-Chat	4.60	4.04	4.32	3.94	3.17	3.56	4.82	3.62	4.22	4.03
SeaLLM-7B-v2.5	6.27	4.96	5.62	5.79	3.82	4.81	6.02	4.02	5.02	5.15
Sailor-14B-Chat	5.26	5.53	5.40	4.62	4.36	4.49	5.31	4.74	5.03	4.97
Qwen2-7B-Instruct	5.93	5.84	5.89	5.47	5.20	5.34	6.17	5.60	5.89	5.70
SeaLLMs-v3-7B-Chat	**6.73**	**6.59**	**6.66**	**6.48**	**5.90**	**6.19**	**6.34**	**5.79**	**6.07**	**6.31**

Table 4: Results of multilingual instruction-following with SeaBench Benchmark

asked in real-life settings, which might be informally written or even ambiguous.

All questions are manually written by native speakers of each language. During construction, we instructed the annotators to ensure the questions were as localized as possible, e.g., using local entities, concepts, and knowledge in the questions. Furthermore, reference answers have been constructed to ensure fair judgment.

Evaluation Details Given the two-turn questions, the model under testing generates two-turn responses in a multi-turn format. These responses are then graded by a stronger LLM (GPT-4o was used in our experiments) using the reference answer to the original questions. The scores are then assigned to each turn of the response.

Results As shown in Table 4, SeaLLMs-v3-7B-Chat outperforms all other models in multilingual instruction-following across Indonesian (id), Thai (th), and Vietnamese (vi). It achieves the highest average scores in both individual turns and overall averages for each language. Specifically, SeaLLMs-v3-7B-Chat surpasses the previous version, SeaLLMs-7B-v2.5, by a significant margin (6.31 vs 5.15) and outperforms the strongest baseline model Qwen2-7B-Instruct (6.31 vs 5.70). These results highlight SeaLLMs-v3-7B-Chat's superior ability to generate more coherent and contextually appropriate multi-turn responses.

4.1.4 Translation

Dataset We evaluate the machine translation performances with the test set of Flores-200 (Costa-jussà et al., 2022). We choose all 12 languages for a comprehensive evaluation, including Burmese (my), Chinese (zh), English (en), Indonesian (id), Javanese (jv), Khmer (km), Lao (lo), Malay (ms), Tagalog (tl), Tamil (ta), Thai (th), and Vietnamese (vi). We translate between each pair of languages and report the average 0-shot chrF scores after averaging the results for target languages.

Results As shown in Table 5, SeaLLMs-v3-7B-Chat outperforms other models in machine

Model	en	id	jv	km	lo	ms	my	ta	th	tl	vi	zh	avg
Sailor-7B-Chat	49.40	49.78	28.33	2.68	6.85	47.75	5.35	18.23	38.92	29.00	41.76	20.87	28.24
SeaLLM-7B-v2.5	**55.09**	**53.71**	18.13	18.09	15.53	**51.33**	19.71	26.10	40.55	**45.58**	**44.56**	24.18	34.38
Meta-Llama-3-8B-Instruct	51.54	49.03	22.46	15.34	5.42	46.72	21.24	**32.09**	35.75	40.80	39.31	14.87	31.22
Qwen2-7B-Instruct	50.36	47.55	29.36	19.26	11.06	42.43	19.33	20.04	36.07	37.91	39.63	22.87	31.32
SeaLLMs-v3-7B-Chat	54.68	52.52	**29.86**	**27.30**	**26.34**	45.04	**21.54**	31.93	**41.52**	38.51	43.78	**26.10**	**36.52**

Table 5: Results of translation with Flores-200.

Model	en	zh	vi	th	id	avg	avg_sea
Gemma-1.1-7B-it	53.61	28.22	26.18	21.28	30.39	31.94	25.95
Sailor-7B-Chat	33.76	18.82	5.19	9.68	16.42	16.78	10.43
SeaLLM-7B-v2.5	13.10	1.53	3.24	19.58	0.78	7.65	7.87
Meta-Llama-3-8B-Instruct	**72.23**	0.00	1.23	0.80	3.91	15.63	1.98
GLM-4-9B-Chat	45.02	40.98	21.48	5.42	2.37	23.05	9.76
Qwen2-7B-Instruct	63.74	35.75	52.86	46.42	55.93	50.94	51.74
SeaLLMs-v3-7B-Chat	71.13	**77.17**	**78.18**	**61.64**	**67.61**	**71.14**	**69.14**

Table 6: Performance in refusing questions about non-existing entities on SeaRefuse-G.

translation, achieving an average chrF score of 36.52. It excels particularly in Javanese, Khmer, Lao, Burmese, Thai, and Chinese, consistently achieving the highest scores in these languages. Compared to its predecessor, SeaLLMs-7B-v2.5, which has an average score of 34.38, SeaLLMs-v3-7B-Chat shows clear improvement. Additionally, SeaLLMs-v3-7B-Chat surpasses strong baselines like Meta-Llama-3-8B-Instruct and Qwen2-7B-Instruct, with average scores of 31.22 and 31.32, respectively. Notably, the model's performance in translating low-resource languages, such as Khmer (27.30) and Lao (26.34), highlights its robustness and effectiveness in handling diverse and challenging translation tasks. This consistent performance across multiple languages underscores the model's versatility and capability in low-resource language translation settings.

4.2 Model Trustworthiness

4.2.1 Hallucination

A trustworthy LLM should only answer the questions that it knows and abstain from answering questions that it does not know. Previous studies reveal that recent LLMs are prone to answering questions that exceed their knowledge boundaries, leading to hallucinated responses (Yang et al., 2023; Zhang et al., 2024). However, evaluating a model's ability to refuse questions it doesn't know is challenging. This requires distinguishing the model's knowledge boundaries, which is difficult since most existing LLMs do not provide transparency in their pre-training data.

To address this challenge, we propose a novel **SeaRefuse** evaluation benchmark, which consists of unanswerable factoid questions about non-existing entities and answerable factoid questions in SEA languages[4]. Unanswerable questions about non-existent entities are designed to surpass the knowledge boundaries of LLMs. Our benchmark includes two test sets: **SeaRefuse-G** and **SeaRefuse-H**. In the SeaRefuse-G test set, the unanswerable questions are generated by prompting GPT-4o. In the SeaRefuse-H test set, our linguists annotate the unanswerable questions by refining machine-generated unanswerable questions. The answerable questions in both SeaRefuse-G and SeaRefuse-H are collected from existing factoid QA datasets including ParaRel (Zhang et al., 2024; Elazar et al., 2021), NLPCC-KBQA (Duan, 2016; Duan and Tang, 2018), and TyDi QA (Clark et al., 2020). For the SeaRefuse-G tese set, each language has 500 answerable and 500 unanswerable questions, except for Vietnamese, which has 483 of each. In the SeaRefuse-H dataset, each language contains 100 answerable and 100 unanswerable questions.

In evaluation, we report the F1-score of each model on correctly refusing questions about non-existing entities. We follow the confusion matrix depicted in Table 8 to compute the F1 scores. We adopt a keyword-matching approach to determine

[4]It will be publicly available at https://huggingface.co/datasets/SeaLLMs/SeaRefuse

Model	en	zh	vi	th	id	avg	avg_sea
Gemma-1.1-7B-it	51.95	29.92	12.07	30.16	35.48	31.92	25.90
Sailor-7B-Chat	33.87	15.79	11.32	12.96	31.67	21.12	18.65
SeaLLM-7B-v2.5	10.91	3.92	11.32	49.66	31.15	21.39	30.71
Meta-Llama-3-8B-Instruct	**73.17**	0.00	0.00	0.00	11.21	16.88	3.74
GLM-4-9B-Chat	35.38	52.50	20.17	5.77	9.52	24.67	11.82
Qwen2-7B-Instruct	58.50	42.75	62.82	60.53	63.51	57.62	62.29
SeaLLMs-v3-7B-Chat	68.54	**81.82**	**83.84**	**84.58**	**89.66**	**81.69**	**86.03**

Table 7: Performance in refusing questions about non-existing entities on SeaRefuse-H.

	Answerable question	**Unanswerable question**
Refused	True-Positive	False-Positive
Answered	False-Negative	True-Negative

Table 8: The confusion matrix for the evaluation of the refusal ability of LLMs.

whether a model refuses to answer a factoid question. Specifically, we work with professional and native linguists to devise a set of refusal keywords for English, Chinese, Vietnamese, Indonesian, and Thai, respectively. If the generated response contains any of these refusal keywords, we assume that the response is a refusal response.

The experiment results on SeaRefuse-G and SeaRefuse-H are shown in Table 6 and Table 7, respectively. We observe that SeaLLMs-v3-7B-Chat outperforms all other baseline models by a large margin in zh, vi, th, and id languages. In English, the performance of SeaLLMs-v3-7B-Chat is competitive with Llama-3-8B-Instruct. These results demonstrate the capability of SeaLLMs-v3 to refuse questions that it does not know.

4.2.2 Safety

Model	en	jv	th	vi	zh	avg
Sailor-7B-Chat	78.7	54.9	62.2	67.6	76.2	67.9
Meta-Llama-3-8B-Instruct	88.3	26.4	71.1	69.8	77.1	66.5
Sailor-14B-Chat	86.9	30.5	53.7	60.9	72.7	60.9
GLM-4-9B-Chat	77.1	21.3	30.2	60.6	74.9	52.8
Qwen2-7B-Instruct	88.6	43.8	63.8	73.0	87.3	71.3
SeaLLMs-v3-7B-Chat	**88.9**	**60.0**	**73.3**	**83.8**	**92.7**	**79.7**

Table 9: Safety performance of different models

To evaluate the models' safety capabilities, we use the questions of SEA languages from the MultiJail dataset (Deng et al., 2023), which includes English (en), Javanese (jv), Thai (th), Vietnamese (vi), and Chinese (zh). Each question in the dataset is potentially malicious, and the model should refuse to answer them. To determine whether the model's response is safe, we first translate the response into English and then prompt GPT-4o to check if the translated response is harmful. The results are reported as the safe rate of the responses.

Table 9 presents the safety capabilities of various models evaluated with the MultiJail dataset. Notably, SeaLLMs-v3-7B-Chat outperforms all other models with an average safe rate of 79.7%, demonstrating robust performance across all languages, particularly excelling in Vietnamese (83.8%) and Chinese (92.7%). In comparison, Qwen2-7B-Instruct follows with a distant second average of 71.3%, with its highest safe rate in Chinese (87.3%). Other models like Sailor-7B-Chat and Llama-3-8B-Instruct also show competitive performance but lag behind in consistency across languages. Notably, the exceptional performance of SeaLLMs-v3 in the three Southeast Asian languages (jv, th, and vi) underscores SeaLLM's effective design, which caters to the linguistic nuances of this region.

5 Conclusion

SeaLLMs 3 represents a significant advancement in the development of large language models for Southeast Asian languages, addressing the region's unique linguistic and cultural challenges. By adopting an efficient language enhancement approach and constructing a comprehensive instruction tuning dataset, SeaLLMs 3 achieves state-of-the-art performance while maintaining cost-effectiveness. Our commitment to reliability and safety, providing contextually appropriate responses, further strengthens the model's applicability and trustworthiness. The open-sourcing of both foundational and chat models ensures that SeaLLMs 3 is accessible for a wide range of applications, fostering further innovation and inclusivity in AI development for Southeast Asia.

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